# Who Decides Washington State?

### Who Returns the Ballot? Age Composition in Washington's Odd-Year Electorate, 2021–2025

**Stephen Kirby** · Tikor Consulting · July 2026

*AI-assisted drafting and analysis review. All figures are reproducible from public-record data available through lawful request and from the open-source scripts cited below, including `scripts/verify_who_decides_wa.py`. The paper source, code, and data-acquisition recipe are public at https://github.com/skirby359/who-decides; the underlying voter file is not redistributed. Contact: kirby@tikorconsulting.com.*

## Abstract

Many of Washington's local offices are filled in odd-year November elections, when turnout runs far below presidential years. This paper asks *who returns those ballots*. Using Washington's statewide voter-registration database, a 5.51-million-voter roll linked to 27.1 million individual vote-history records with the voter's year of birth, it measures the age make-up of every November general electorate from 2021 through 2025. The central finding is descriptive: Washington's odd-year electorate is markedly older than its presidential one. Voters 65 and older were 36.7%, 40.2%, and 40.3% of the 2021, 2023, and 2025 odd-year electorates, against 28.5% in 2024; voters 18–29 fell from 14.2% in 2024 to about 7.6% off-cycle. The result survives voter-file coverage validation against certified ballot counts, formal bounds for current-roll survivorship, a closer-in-time roll cross-check, alternative birth-year assumptions, county-level checks, and exclusion of any single off-year. The off-year electorate is also older than the registered roll and the citizen voting-age population. Individual records show it is largely the presidential electorate's *habitual core* (92–97% of off-year voters also vote in presidential years), while the peripheral voters who drop off off-cycle skew young. The paper measures ballot return, not votes cast in specific down-ballot contests, and does not estimate partisan or policy consequences. Its contribution is a validated, individual-record measurement of age composition across the presidential–midterm–off-year salience gradient, in a universal vote-by-mail state where the formal administrative cost of voting is comparatively low.

**Keywords:** election timing; voter turnout; local elections; age representation; voter files; Washington; vote by mail; off-cycle elections.

---

## The question

Coverage of "turnout" usually asks how *many* people vote. The question that matters more for how a state is governed is *who*, and the answer shifts depending on which election you look at. Washington fills many of its most local offices such as city councils, school boards, port and fire commissions, and many county and judicial seats in **odd-year, off-cycle Novembers** (RCW 29A.04.321 limits the odd-year statewide general election chiefly to city, town, and district offices, certain county positions, and state measures), when only about **38%** of registered voters return a ballot. This paper looks at who that 38% is, voter by voter.

The short answer: **the odd-year electorate we can observe is markedly older than the presidential one.** It is not a smaller copy of the presidential electorate; rather, it is an older one.

**What is measured.** This paper measures **ballot return**, that is whether a registered voter cast a November ballot, one voter at a time. It does *not* tell us whether that voter actually marked a specific city-council, school-board, port, fire, judicial, or county race; the voter file records that someone voted in an *election*, not in a given *contest* (Lucero et al. 2025). That gap matters most for the **even-year what-if**: if local races were moved onto the longer even-year ballot, some of the added voters would return a ballot but skip the local race. For the **odd-year electorate described here** the gap is smaller. Washington's odd-year ballot is mostly local and district contests, with only the occasional statewide measure so ballot return is a closer stand-in for local-race participation, though it is still an election-level measure, not a contest-level one. Appendix F takes up roll-off as the live counterargument.

---

## Data and validation

**Source.** `voting_history` (27.1M records) joined to `voters` (year of birth, ~100% coverage) from Washington's standard statewide VRDB extract (April 2026; provenance and use terms in Appendix B). Age is based on election year − birth year; cohorts are assigned per election. November generals only, 2021–2025.

The sharpest objection to any voter-file study is: *you did not measure the electorate; you measured the part of it still visible in your current voter file*. Because the vote-history table is tied to a current (2026) roll, anyone who voted in 2021–2024 but has since died, moved, or been dropped is missing. If those departed voters were a random slice, losing them would not matter; if they skew young, the finding would be overstated. Both are testable, and the answers favor the finding.

**Coverage is high, and the worst case is bounded directly.** We benchmark the reconstruction against certified WA Secretary of State ballot counts, from the per-election turnout pages ( `results.vote.wa.gov/results/<yyyymmdd>/turnout.html` ; e.g., November 2021: 1,896,481 ballots counted, 39.38% turnout). "In file" counts distinct voters in the cumulative vote-history table for an election; "Analyzable" also requires a match to the April 2026 roll with a year of birth. Analyzable coverage runs from **90.8% in 2021 to 99.6% in 2025**, improving sharply over time. The weakest year is the 2021 off-year, which is exactly why the analysis reports an explicit worst-case bound (below) instead of assuming the missing voters look like the ones we can see. The benchmark is *certified ballots counted*; the vote-history file is a voter-level reconstruction of who got ballot credit, expected to track that count closely but not to match it exactly.

| Election | Type | Official ballots counted | In file | Analyzable (roll + YOB) | Analyzable / official |
|---|---|---|---|---|---|
| Nov 2021 | Off-year | 1,896,481 | 1,828,231 (96.4%) | 1,722,262 | **90.8%** |
| Nov 2022 | Midterm | 3,067,686 | 3,025,352 (98.6%) | 2,884,966 | **94.0%** |
| Nov 2023 | Off-year | 1,758,084 | 1,731,431 (98.5%) | 1,686,656 | **95.9%** |
| Nov 2024 | Presidential | 3,961,569 | 3,958,965 (99.9%) | 3,880,070 | **97.9%** |
| Nov 2025 | Off-year | 2,001,425 | 1,995,509 (99.7%) | 1,993,505 | **99.6%** |

**The voters we lose skew old, and the rest is bounded.** A second roll snapshot ( `voters_20230901` , 5.29M rows, Sept 2023) lets us partly describe voters who cast a past ballot but are gone from the April 2026 roll. Among these visible drop-offs, seniors are heavily overrepresented:

| Missing-from-current-roll voters | 65+ share | 18–29 share |
|---|---|---|
| Cast a ballot Nov 2021 (n≈106K) | **60.4%** | 6.8% |
| Cast a ballot Nov 2022 (n≈140K) | **60.3%** | 7.1% |
| Cast a ballot Nov 2023 (n≈45K) | **69.0%** | 4.5% |

Leaving the roll is age-loaded more broadly: of the 504K voters (9.5%) who left the roll between 2023 and 2026, 33.1% were 65+, versus 23.9% of those who stayed which is consistent with mortality and age-related moves and cancellations. So the current-roll reconstruction **likely understates** the senior share of past electorates. But since not every missing ballot can be pinned down from the second snapshot, we also report a **formal bound** that assumes nothing about the rest: the 65+ share of the *full certified electorate* first if every unobserved voter (official − analyzable) were under 65, then if every one were 65+.

| Off-year 65+ share of the certified electorate | all missing < 65 (min) | observed | all missing 65+ (max) |
|---|---|---|---|
| Nov 2021 (residual 9.2%) | **33.4%** | 36.8% | 42.6% |
| Nov 2023 (residual 4.1%) | **38.6%** | 40.2% | 42.6% |
| Nov 2025 (residual 0.4%) | **40.2%** | 40.3% | 40.6% |

"Observed" is the 65+ share of the analyzable electorate from the validation table (2021 = 36.8%; the composition table's 36.7% adds a "registered on/before the election" filter, a 0.1-point difference). Min/max apply the extreme assumption to the residual = official − analyzable.

Even under the most hostile assumption, in which every unobserved off-year ballot was cast by someone under 65, each off-year electorate (**33.4% / 38.6% / 40.2%** 65+) stays older than the presidential electorate under *its* most favorable assumption (**≤30.0%**, i.e. even if every missing 2024 ballot were 65+). The conclusion is that the finding does not depend on how the missing voters are assigned. Two points follow: given the attrition evidence, the observed estimates are **likely conservative**, and the **formal worst-case bound** keeps the finding either way.

**A direct check: the composition barely moves under a closer-in-time roll.** The bound above says nothing about the missing residual; a separate check asks whether reconstructing from a *current* roll skews the composition at all. The database's Sept-2023 snapshot ( `voters_20230901` ) still holds voters who have since dropped off the 2026 roll, so it rebuilds the 2021–2023 electorates from a roll much closer to those elections. The two reconstructions agree to within ~1.4 points, and what movement there is runs *upward*, consistent with the current roll slightly undercounting seniors:

| Election | Current-roll 65+ | Sept-2023-snapshot 65+ | Δ |
|---|---:|---:|---:|
| Nov 2021 (off-year) | 36.8% | 38.1% | +1.4 |
| Nov 2022 (midterm) | 31.0% | 32.4% | +1.4 |
| Nov 2023 (off-year) | 40.2% | 41.1% | +0.9 |

The snapshot, which recovers older voters the current roll has since dropped, comes out slightly *higher*, not lower, so the composition finding is not an artifact of building it from a current roll. The 2021 and 2022 comparisons are the cleanest, because the September-2023 snapshot comes after those elections. Read the 2023 comparison more cautiously: the snapshot predates the November-2023 election by two months, so it misses late-2023 registrants who voted that November, and if those late registrants skew younger, as new registrants often do, the snapshot's 2023 estimate may run slightly *high*. Even so, the check reassures: the closer-in-time reconstruction never makes an electorate younger than the current-roll version, and 2021–2022 point the same way.

---

## What the data shows

The robust cut is **composition**: each age group's share of the ballot-returning electorate. It needs no turnout denominator, rests on three consistent off-year cycles, and (per the validation above) survives an explicit worst-case bound.

| Election | Type | 18–29 | 30–44 | 45–64 | 65+ |
|---|---|---:|---:|---:|---:|
| Nov 2024 | Presidential | 14.2% | 24.9% | 32.4% | 28.5% |
| Nov 2022 | Midterm | 10.4% | 22.9% | 35.7% | 31.0% |
| Nov 2021 | Off-year | 7.5% | 19.7% | 36.0% | 36.7% |
| Nov 2023 | Off-year | 7.4% | 19.2% | 33.2% | 40.2% |
| Nov 2025 | Off-year | 8.0% | 19.9% | 31.7% | 40.3% |

- **The off-year electorate is a senior-plurality electorate, and stably so.** Voters 65+ make up **~37–40%** of it (36.7 / 40.2 / 40.3% across 2021 / 2023 / 2025) versus **28.5%** in the presidential year; the 18–29 share falls from **14.2%** to **~7.6%**.
- **The senior-to-youth ratio roughly triples off-cycle** from about **2:1** in the presidential year to **~5:1** off-year with the midterm in between (31.0% 65+). The *off-year* figure is stable across three cycles; the presidential and midterm points are single elections (2024, 2022), so read the ordering as consistent with a salience gradient, not as a smoothly estimated curve.

**Residents, registrants, voters: a rising age ladder.** The off-year electorate is older than the presidential one, older than the registered roll, older than the citizen voting-age population, and older than all adult residents. Lining the ballot-returning electorates up against the roll (April 2026), the citizen voting-age population, and all adult residents (both ACS 2020–24) gives a ladder that climbs at every rung:

| Population | 18–29 | 30–44 | 45–64 | 65+ | Median age |
|---|---:|---:|---:|---:|---:|
| WA adult residents (ACS 2020–24) | 20.0% | 28.3% | 30.5% | 21.1% | ~46† |
| WA citizen voting-age population (ACS 2020–24) | 19.8% | 26.7% | 30.9% | 22.6% | ~47† |

| Population | 18–29 | 30–44 | 45–64 | 65+ | Median age |
|---|---|---|---|---|---|
| Registered roll (April 2026) | 16.7% | 27.2% | 29.8% | 26.3% | 48 |
| 2024 presidential ballot-returners | 14.2% | 24.9% | 32.4% | 28.5% | 52 |
| Off-year ballot-returners (2021/23/25 avg) | 7.6% | 19.6% | 33.6% | 39.1% | 59 |

† Age composition from ACS 2020–24 five-year (the most recent five-year vintage; the 2024 one-year is consistent), reproduced by `scripts/acs_wa_adult_age.py`: total 18+ residents from table B01001, and the **citizen voting-age population (CVAP)** (the eligible-electorate benchmark, which excludes non-citizens) from table B29001. Median ages for the two ACS rows are interpolated from the age brackets and the roll and ballot-returner medians are computed from year of birth and are integer-year (±1) approximations. Roll and ballot-returner figures from the VRDB (`scripts/verify_who_decides_wa.py` #23).

The 65+ share climbs **21.1% → 22.6% → 26.3% → 28.5% → 39.1%** across the five rows; the 18–29 share falls **20.0% → 19.8% → 16.7% → 14.2% → 7.6%**. The median off-year ballot-returner is **59**, about a decade older than the median registered voter (48), and more than a decade older than the median citizen-voting-age adult (~47). Residents, eligible citizens, registrants, and voters are four different populations, and what moves the composition is participation, not the roll. The roll's senior share is low and steady (26.3% on the full April 2026 roll; ~22–25% on the per-election eligible roll in these years) while the off-year returner share reaches ~40%.

**One number for "how unrepresentative."** We can collapse the ladder into a single dissimilarity index, that is how far each electorate's age distribution sits from the citizen voting-age population, taken as half the summed absolute differences across cohorts, where 0 means identical. It comes out **7.4** for the 2024 presidential electorate, **13.2** at the midterm, and **18.5–19.9** across the three off-years. The off-year electorate is roughly **2.5× as age-unrepresentative** of the eligible population as the presidential one. The index depends on the age bins chosen, so treat its value here as comparative, not absolute.

Recorded gender shows a much smaller secondary pattern: the electorate is majority recorded-female throughout, rising from 52.5% in the 2024 presidential electorate to 53.0–53.1% in the off-year electorates, concentrated among older voters. Because the shift is small and administrative gender is not the paper's focus, the analysis treats age as the primary dimension.

The mechanism is differential participation. The within-cohort rates below make it concrete. 18–29 year old participation falls from **58.4%** (2024) to about **16%** off-year, while 65+ slips only from **88.3%** to **~61%**. These rates *back up* the composition finding; they do not carry it:

**Within-cohort participation rate (current-roll reconstruction, not official turnout):**

| Election | Type | 18–29 | 30–44 | 45–64 | 65+ | All |
|---|---|---|---|---|---|---|
| Nov 2024 | Presidential | 58.4% | 68.6% | 80.1% | 88.3% | 75.0% |
| Nov 2022 | Midterm | 36.4% | 52.5% | 69.6% | 84.6% | 62.4% |
| Nov 2021 | Off-year | 16.5% | 28.6% | 43.1% | 65.6% | 39.4% |
| Nov 2023 | Off-year | 14.5% | 24.6% | 37.1% | 59.0% | 34.9% |
| Nov 2025 | Off-year | 16.4% | 27.0% | 39.0% | 59.3% | 36.7% |

**These are not official turnout rates.** The denominator is the age-eligible **April 2026 roll**, not the roll as it stood on each election day, so a later (larger) roll mechanically pulls them down: the "All" column (e.g., 75.0% in 2024) generally sits below Washington's official general-election turnout (39.38% 2021, 63.82% 2022, 36.41% 2023, 78.95% 2024, 39.24% 2025); the exception is 2021, where the reconstruction lands at about the official 39.38%. Read the table as a current-roll *reconstruction* of within-cohort participation. It also rests on a single presidential (2024) and single midterm (2022) cycle.

---

## Sensitivity

The finding does not hinge on the cohort boundaries, the birth-year assumption, any single off-year, or King County.

**Finer cohorts.** Splitting the endpoints sharpens the pattern: the 75+ share rises from **11.8%** in the presidential year to **16.8–18.3%** off-year, while the 18–24 share falls from **7.7%** to **~3.7–4.0%**.

| Election | Type | 18–24 | 25–29 | 30–44 | 45–64 | 65–74 | 75+ |
|---|---|---|---|---|---|---|---|
| Nov 2024 | Presidential | 7.7% | 6.5% | 24.9% | 32.4% | 16.7% | 11.8% |
| Nov 2022 | Midterm | 5.3% | 5.1% | 22.9% | 35.7% | 19.4% | 11.7% |
| Nov 2021 | Off-year | 3.7% | 3.8% | 19.7% | 36.0% | 23.3% | 13.4% |
| Nov 2023 | Off-year | 3.7% | 3.6% | 19.2% | 33.2% | 23.4% | 16.8% |
| Nov 2025 | Off-year | 4.0% | 4.0% | 19.9% | 31.7% | 22.1% | 18.3% |

**Birth-year assumption.** The file gives year of birth, not the full date, so the main analysis takes age = election year − birth year to standardize age (We assume the birthday has already happened by the November election). That can label voters with late-November or December birthdays as a year older than they really were on Election Day. As a check, we recompute the 65+ share under the opposite extreme, treating every voter as if their birthday had *not* yet come (a Dec-31 assumption). That moves the off-year 65+ share by **≤2.4 points** (e.g., 2021: 36.8% → 34.3%; 2025: 40.3% → 38.2%) and leaves the presidential/off-year gap intact (the presidential share moves too, 28.5% → 26.7%). Because November falls late in the year, the true value should sit *closer to the main convention* than to this all-younger extreme, barring unusual late-year birthday clustering.

**Off-year stability, and statewide measures.** The three off-years land on the same result (65+ share 36.7 / 40.2 / 40.3%) despite *different* statewide ballot content: 2021 and 2023 carried only the state's non-binding tax "advisory votes," and 2025's only statewide item was a single fiscal constitutional amendment (SJR 8201, on investing the WA Cares trust fund; approved per the WA Secretary of State's certified November 4, 2025 results, `results.vote.wa.gov`, where it was the only statewide contest). None is a high-salience, mobilizing contest, and the fact that the 65+ share barely moves across all three is direct evidence that none of them drives the composition. Dropping any one off-year leaves the conclusion intact.

**Geography.** King County (the largest and youngest) runs younger than the rest at every salience level, but the gradient shows up everywhere, and is steeper outside the urban core.

| 65+ share of electorate | 2024 Pres | 2021 Off | 2023 Off | 2025 Off |
|---|---|---|---|---|
| King County | 23.0% | 28.7% | 32.2% | 30.7% |
| Rest of state | 30.7% | 40.4% | 43.5% | 45.0% |
| Metro counties[1] | 26.4% | 34.3% | 37.5% | 37.3% |
| Rural counties | 37.7% | 46.7% | 51.5% | 54.2% |

[1] Metro = the ten most-populous counties (King, Pierce, Snohomish, Clark, Spokane, Thurston, Kitsap, Whatcom, Benton, Yakima); rural = the remaining 29.

The presidential→off-year shift toward seniors is positive in **all 39 counties** (full breakdown in Appendix E).

---

## Interpretation: mechanism, lever, and policy caution

**It is behavior, not the rolls.** What changed the 65+ share off-cycle? Are older people turning out at higher rates, or is the registration roll itself getting older? A standard decomposition that separates the two (Kitagawa–Das Gupta; `scripts/diag_turnout_decomposition.py`) puts it almost entirely on behavior. Of the **+11.8-point** rise in the 65+ share from 2024 to 2025, **+10.9 points (92%) come from turnout rates** and only **+0.9 from a changing roll**; for 18–29 the split is **−6.0 of −6.2 points (97%) behavioral.** The pattern holds across all three off-years as the turnout accounts for **92% (2025), 95% (2023), and 79% (2021)** of the 65+ rise and in 2021 and 2023 the roll effect is actually slightly *negative* (the roll was marginally younger, so behavior more than accounts for the shift). This is robust to the survivorship worry: both years in each comparison are read off nearly the same recent roll, so any current-roll distortion is shared and largely cancels, and correcting the older-skewing attrition described earlier would only *raise* the senior turnout rate, not move weight onto the roll. The skew is a turnout-and-salience story, which is why the lever is **when you hold the election**, not registration policy.

**The off-year electorate is the presidential electorate's habitual core.** Because we can follow individual voters, we can see surge-and-decline directly rather than infer it from aggregates. Roughly **92–97%** of each off-year electorate also cast a 2024 presidential ballot, but only **42–48%** of 2024 presidential voters showed up in a given off-year, so the off-year electorate is close to a *standing core* of the presidential one. Sorting 2024 presidential voters by whether they *also* voted in the 2023 off-year shows who stays and who drops off: the **habitual core** (voted both; 1.6M) is **42.8% 65+ and 6.1% under 30**, while the **presidential-only group** (2.2M) is **18.0% 65+ and 20.2% under 30**. The voters who mostly turn up when a presidential race is on the ballot are disproportionately young, and the off-year electorate is what is left once they fall away. Off-year voters have also been registered longer, a median of about **16–17 years** since they registered, versus **12** for the presidential electorate. How long someone has been registered is only a rough proxy for lifelong civic attachment, and because this comparison is limited to voters still on the current roll, it should be read next to the survivorship checks above, in which attrition drops some past voters from the panel, especially older ones who died or moved before the 2024 comparison point.

**The lever.** The evidence that on-cycle timing reshapes the electorate is strong, and much of it is quasi-experimental. Anzia (2014) and Hajnal & Trounstine (2005) show that off-cycle elections shrink the electorate and make it less representative, tilting outcomes toward organized, high-turnout groups. Hajnal, Kogan & Markarian (2022), using individual micro-targeting data, find that California's shift to on-cycle municipal elections roughly **doubles** local turnout and makes the electorate considerably more representative by age, race, and partisanship. Lucero et al. (2025), surveying cities that switched, put voters over 45 at **58.4%** of the off-cycle electorate versus **49.7%** of the presidential-year one (and, citing Hajnal et al., a roughly **22-point** over-55 gap). This paper measures Washington's *gap* (about 40% 65+ off-year versus about 28.5% presidential) not what consolidation would actually do here. But **if Washington behaved like the places those studies cover, moving local races onto even-year Novembers would produce a substantially larger and younger local electorate**. A reasonable extrapolation from California's experience, not a Washington simulation.

**A caution on policy.** There is good reason to *expect* that a smaller, older, more-organized off-cycle electorate yields different policy (Anzia 2014, on public-employee pay; Kogan, Lavertu & Peskowitz 2018, on school-district spending). But the evidence is not settled. **Ornstein (2024)**, looking at California's 2018 on-cycle mandate (SB 415) across 236 local governments, finds the expected gains in turnout and diversity but **no** detectable knock-on effect on who gets represented, who runs, the incumbency advantage, housing policy, or public-employee salaries. **This paper does not try to settle that debate.** It measures participation and composition, and the composition result is not, by itself, evidence that policy is being captured. (The full list of objections is in Appendix A: preference-intensity, ballot dilution / roll-off, and age-as-proxy.)

---

## What this paper does not claim, and limits

- **No partisan consequence.** Washington does not publish party of record, so this paper stops at participation and composition. The party-registration companions (New York, Idaho) test that.
- **No policy-outcome claim.** The composition finding is not evidence of policy capture, union influence, or changed local outcomes; the literature is mixed (above) and it is not tested here.
- **No contest-level claim.** This paper measures ballot return in November, not participation in a specific local contest (Lucero et al. 2025). Appendix F estimates even-year *contest*-level roll-off as a first look at the sequel.
- **No individual-level causal claim** about why any person votes or abstains.
- **Age vs cohort.** Five cycles (2021–2025) cannot separate a *life-cycle* effect (people vote off-cycle more as they age) from a *cohort* effect (a durable high-turnout generation that happens to be old right now). The habitual-core result fits either; telling them apart needs a longer individual panel. Appendix H's smooth single-year-of-age curve is consistent with the life-cycle reading but is subject to the same limit.
- **Cycle coverage.** The vote history begins in 2021, so the presidential row rests on **2024 alone** and the midterm on **2022 alone**; only the off-year row averages three cycles. King County 2020 presidential is not loaded and is excluded (all figures are 2021+).
- **Uncertainty.** The VRDB figures are near-complete counts and carry no sampling error; the ACS resident/CVAP rows are 5-year *estimates* with margins of error (small at this geography and level of aggregation, but not zero); and the county roll-off correlation (n=39) is reported without a confidence interval; it is descriptive, not inferential.
- **The rates are a current-roll reconstruction**, not official turnout (above); the **share-of-electorate** figures, which need no denominator and are bounded and validated, are what carry the finding. Methods detail in Appendix C.

---

# Appendices

## Appendix A — The objections, in full

The most obvious objection is also the weakest: **this is voluntary participation, not disenfranchisement.** Washington votes entirely by mail, postage prepaid, with automatic and same-day registration, so no one is shut out of the off-year ballot, and the fix, moving local races onto even-year ballots, is an ordinary scheduling choice. This is a question of **design**, not rights. Explaining *why* the off-year electorate is older does not make it any less old: the gap is there however you judge it. (It is also what makes Washington a useful test case. In a prepaid, same-day-registration state the usual "too hard to vote" explanation for young drop-off is weak, so the age gap is hard to pin on friction.)

Three other objections:

**1. Maybe the off-year electorate is a filter for engagement, not a defect.** Perhaps the people who bother to return a low-salience local ballot are better informed, more rooted locally, and more directly affected by property taxes, schools, and utility districts, while presidential-year voters, pulled in by the national race, know *less* about local contests and lean on national cues. This is the real normative crux, and three things bound it. (a) It is a claim about the *quality* of the marginal voter, which a participation study like this one cannot measure or settle. (b) It runs against a large body of work finding that on-cycle electorates are *more* representative of the population by age, race, and partisanship (Hajnal, Kogan & Markarian 2022), closer to the community the government actually serves. (c) The information gap it assumes is itself partly a product of off-cycle timing (thin news coverage), so it is not cleanly separate from the scheduling choice.

**2. Ballot dilution / down-ballot roll-off.** If local races move to even years, more people get the ballot but some skip the local contest, so the number actually voting in that race grows by less than total turnout does. Appendix F measures this directly for Washington. In the 2024 even-year general, roll-off, the share of returned ballots that skip a given contest, was **~3–7%** for partisan statewide offices and ballot measures, **~16–17%** for *contested* nonpartisan statewide races (Supreme Court, Superintendent of Public Instruction), and **~34%** for *uncontested* ones. Nonpartisan judicial races are the closest even-year stand-in for the local nonpartisan races (city council, school board) that consolidation would add, and probably an upper bound, since voters know least about judges. That is higher than the classic ~2–10% estimate (Wattenberg, McAllister & Salvanto 2000, who tie roll-off largely to *information*), and we do not assume it is age-neutral for Washington's local races, as that is untested. **Even so, the net effect is almost certainly a bigger electorate**, though how much bigger depends on whether local races land on a presidential or a midterm-year ballot (Appendix F). Even at the worst roll-off observed (34%), the electorate actually deciding the race is ~52% of registered voters on a presidential ballot and ~42% on a midterm one. In either case the deciding electorate remains above the ~38% who turn out off-cycle today. The gain survives heavy roll-off unless local roll-off runs far above the contested-nonpartisan mark.

**3. Age is not a clean proxy for whose interests matter.** Seniors are heavily affected by local taxes, emergency services, transit, utilities, public safety, housing supply, and school levies, and this paper makes no claim that younger people's interests count for more. The narrower claim is about **representation**, and it stays entirely inside the registered electorate the data covers: among registered voters, off-year ballot-returners are much older than presidential-year ones (median age ~59–60 vs 52, and about a decade older than the median registrant at 48). If a markedly older slice of the registered electorate is choosing local officials, then the preferences recorded at the ballot box differ systematically by age. The off-year electorate is ~39% 65+, over **1.7 times** the **22.6%** of citizen voting-age Washingtonians who are 65+ (ACS 2020–24 CVAP) and nearly double the 21.1% of all adult residents, while its 18–29 share (~7.6%) is under two-fifths of the ~20% in both benchmarks (§ What the data shows). Whether that is desirable, tolerable, or a problem is a normative question; what this paper contributes is a measurement of its size.

## Appendix B — Data access and privacy

- **What was obtained.** Washington's standard statewide **VRDB extract**, the single public extract the Secretary of State publishes (registrants plus cumulative vote history), regenerated monthly. This study uses the **April 2026 extract** (requested April 8, 2026). By statute, the public file is limited to each voter's name, address, political jurisdiction, gender, **year of birth**, voting record, registration date, and registration number, and **no other information from voter-registration records is available for public inspection or copying** (RCW 29A.08.710). This is the statutory basis for using year of birth rather than full date of birth. Its use is restricted to elections and political purposes and **may not be used for commercial purposes** (RCW 29A.08.720); the file is **not redistributable**, with penalties at RCW 29A.08.740. Full provenance and access date: Data Sources & Reproducibility.
- **Year of birth, not date of birth.** Consistent with the statute, the extract supplies **year of birth only**; full date of birth is protected information that Washington withholds from the public voter database, and the legislature further strengthened voter-data protections in 2026 (SB 5892 / Ch. 213, Laws of 2026, eff. Mar. 25, 2026). In our file every birth value resolves to a July-1 sentinel (a storage placeholder marking year-only granularity, not a birthday; ages in this paper are computed from birth year alone, as described in Section II), confirming that **no full date of birth was obtained, stored, or used.**

- **What is released.** Only aggregate cohort counts, with cell sizes in the thousands to millions; no individual-level records, addresses, or names. The analysis scripts emit aggregates only, and the repository's product layer additionally enforces a PII firewall ( `src/wa_analyzer/product/firewall.py` ). Only **citations and code, not data** are published.

## Appendix C — Methods

- **Source and unit.** `voting_history` (27.1M records) joined to `voters` (year of birth, ~100% coverage), cohorts assigned per election. November generals only. The unit is **ballot return**, not participation in a specific contest.
- **Age convention.** With year of birth, age = election year − birth year (equivalently, birthday assumed reached by November). This is a ±1-year approximation of true age; the birth-year-imputation sensitivity (Sensitivity) shows the alternative extreme moves the off-year 65+ share by ≤2.4 points and leaves the gap intact.
- **Composition vs rates.** The within-cohort *participation rates* are a current-roll reconstruction (denominator = the April 2026 roll, not the election-day registered cohort), so they are not official turnout and are reported only to show the mechanism. The **share-of-electorate** figures, which need no denominator, carry the finding.
- **Coverage and bounding.** Coverage of the analyzable electorate against certified counts is 90.8–99.6%. Two distinct claims: the observable attrition component skews old, so the observed estimates are *likely conservative*; and, assuming nothing about the residual, a formal worst-case bound (every missing voter under 65) still keeps each off-year 65+ share above the presidential one, so the finding does not depend on the residual's composition.
- **Decomposition.** The behavior-vs-rolls split is the symmetric two-factor (Kitagawa–Das Gupta) standardization (Appendix D).
- **Reproduction.** `scripts/verify_who_decides_wa.py` re-derives every count in the tables above from scratch (validation, survivorship, bounding, finer cohorts, imputation, geography, decomposition, habitual-core overlap, snapshot cross-validation, gender, representativeness index, single-year-of-age curve), independently of the analysis code; the ecological roll-off correlation is in `scripts/diag_wa_rolloff_2024.py`, and the precinct-level SES-controlled sequel (Appendix F) in `scripts/diag_wa_rolloff_precinct.py`.

## Appendix D — Related work

The core finding, that low-salience electorates are older and smaller, is well established, and this paper does not claim to discover it. The contribution is narrower, and twofold. First, it gives a **validated, individual-record** measurement of Washington's recent November electorates across the full salience gradient (presidential → midterm → off-year), from ~100% of the state's vote records, in a **universal vote-by-mail, same-day-registration** state where the usual "too hard to vote" explanation for youth drop-off is weak. Second, and just as important, it tackles a common voter-file problem head-on: because past vote records are reconstructed from a *current* registration file, voters who have since died, moved, or been dropped can go missing, an assumption many voter-file studies leave unstated. By benchmarking against certified ballot counts, describing the attrition we can see, and formally bounding the rest, the paper shows the age-composition result is not an artifact of current-roll survivorship.

- **Turnout composition by salience (surge-and-decline).** Campbell, "Surge and Decline: A Study of Electoral Change," *Public Opinion Quarterly* 24(3) (1960): 397–418. Wolfinger & Rosenstone, *Who Votes?* (Yale, 1980); Leighley & Nagler, *Who Votes Now?* (Princeton, 2013); age is among the most durable turnout predictors. Plutzer, "Becoming a Habitual Voter: Inertia, Resources, and Growth in Young Adulthood," *American Political Science Review* 96(1) (2002): 41–56; turnout is a habit that develops over the life cycle, the frame Appendix H's single-year-of-age curve is consistent with.
- **Off-cycle election timing, composition, and representation.** Anzia, *Timing and Turnout: How Off-Cycle Elections Favor Organized Groups* (Univ. of Chicago Press, 2014); Hajnal & Trounstine, "Where Turnout Matters," *Journal of Politics* 67(2) (2005): 515–535; Hajnal, Kogan & Markarian, "Who Votes: City Election Timing and Voter Composition," *American Political Science Review* 116(1) (2022): 374–383; Kogan, Lavertu & Peskowitz, "Election Timing, Electorate Composition, and Policy Outcomes: Evidence from School Districts," *American Journal of Political Science* 62(3) (2018): 637–651; Lucero, Robles, Trounstine & Collins, "What Date Works Best for You? Changes in Electorate Demographics and Policy Priorities in Concurrent Elections," *Urban Affairs Review* (2025); Einstein, Palmer, Hamilton & Singer, "Age and Homeownership Drive the Local Turnout Gap," *Urban Affairs Review* (2025); the last is the closest analog to the age result here.
- **Contested policy effects of timing.** Ornstein, "Election Timing Revisited: Evidence from California's Voter Participation Rights Act" (working paper, 2024); turnout/diversity gains but null downstream policy effects.
- **Down-ballot roll-off.** Wattenberg, McAllister & Salvanto, "How Voting Is Like Taking an SAT Test: An Analysis of

American Voter Rolloff," *American Politics Quarterly* 28(2) (2000): 234–250, doi:10.1177/1532673X00028002005; roll-off is substantially information-driven.

- **Voter-file / individual-level method.** Ansolabehere & Hersh, "Validation: What Big Data Reveal About Survey Misreporting and the Real Electorate," *Political Analysis* 20(4) (2012): 437–459; Hersh, *Hacking the Electorate* (Cambridge, 2015).
- **Decomposition method.** Kitagawa, "Components of a Difference Between Two Rates," *JASA* 50(272) (1955): 1168–1194; Das Gupta, *Standardization and Decomposition of Rates* (U.S. Census Bureau, P23-186, 1993).

## Appendix E — 65+ share of the electorate by county

The off-cycle senior tilt is not a King County artifact or a rural artifact: the presidential→off-year shift toward seniors is **positive in all 39 counties** ( `scripts/verify_who_decides_wa.py` #24). Counties are sorted by their 2024 presidential 65+ share; the last column is the average off-year (2023, 2025) share minus the presidential share. The off-year average uses 2023 and 2025 because their analyzable coverage is much higher than 2021's (99.6% and 95.9% vs 90.8%); adding 2021 does not change the conclusion: averaged over all three off-years the gap stays positive in every county (King +7.5 to Franklin +16.1).

| County | 2024 Pres | 2023 Off | 2025 Off | Pres→Off | County | 2024 Pres | 2023 Off | 2025 Off | Pres→Off |
|---|---|---|---|---|---|---|---|---|---|
| Jefferson | 53.2% | 66.1% | 65.7% | +12.7 | Douglas | 34.3% | 50.8% | 53.2% | +17.7 |
| Pacific | 48.2% | 60.1% | 62.3% | +13.0 | Cowlitz | 32.5% | 46.9% | 49.7% | +15.8 |
| Clallam | 47.3% | 59.9% | 60.9% | +13.1 | Kitsap | 32.2% | 44.1% | 44.5% | +12.1 |
| San Juan | 47.3% | 59.5% | 57.8% | +11.4 | Grant | 32.0% | 47.1% | 48.3% | +15.7 |
| Ferry | 45.4% | 55.7% | 62.1% | +13.5 | Yakima | 31.5% | 47.4% | 48.1% | +16.3 |
| Wahkiakum | 45.0% | 54.0% | 56.9% | +10.5 | Klickitat | 31.3% | 45.0% | 44.5% | +13.4 |
| Island | 43.0% | 54.5% | 60.1% | +14.3 | Thurston | 30.9% | 43.4% | 42.8% | +12.2 |
| Columbia | 42.4% | 51.8% | 55.9% | +11.5 | Adams | 30.2% | 43.2% | 43.5% | +13.1 |
| Garfield | 42.4% | 54.4% | 50.5% | +10.1 | Whatcom | 29.6% | 37.5% | 40.0% | +9.1 |
| Asotin | 40.9% | 57.8% | 58.3% | +17.1 | Spokane | 29.6% | 40.3% | 43.2% | +12.2 |
| Okanogan | 40.9% | 50.2% | 55.1% | +11.8 | Benton | 29.1% | 42.4% | 44.9% | +14.6 |
| Mason | 40.0% | 55.7% | 55.7% | +15.7 | Clark | 27.9% | 44.2% | 42.3% | +15.4 |
| Pend Oreille | 39.9% | 54.8% | 58.6% | +16.8 | Pierce | 26.7% | 39.9% | 40.4% | +13.4 |
| Lincoln | 39.6% | 51.4% | 54.2% | +13.2 | Whitman | 26.3% | 38.5% | 40.4% | +13.1 |
| Grays Harbor | 39.4% | 54.7% | 57.6% | +16.8 | Snohomish | 25.5% | 36.8% | 38.4% | +12.1 |
| Kittitas | 38.6% | 49.8% | 54.1% | +13.3 | Franklin | 24.0% | 39.8% | 42.6% | +17.2 |
| Stevens | 38.4% | 50.3% | 56.0% | +14.7 | King | 23.0% | 32.2% | 30.7% | +8.4 |
| Skagit | 38.3% | 52.3% | 54.3% | +15.0 | | | | | |
| Walla Walla | 35.6% | 48.6% | 52.3% | +14.8 | | | | | |
| Chelan | 35.2% | 45.2% | 51.6% | +13.2 | | | | | |
| Lewis | 34.7% | 50.0% | 51.9% | +16.2 | | | | | |
| Skamania | 34.5% | 48.9% | 49.8% | +14.9 | | | | | |

The off-year 65+ share ranges from **30.7% (King)** to **66% (Jefferson)**, and every county moves toward seniors off-cycle (gaps +8.4 to +17.7 points). The result is a statewide phenomenon; King is simply its youngest instance.

## Appendix F — Contest-level roll-off on the even-year ballot

The one thing this paper does *not* measure is whether a voter marked a specific contest. The obvious sequel, and the live objection to consolidation, is **down-ballot roll-off**: if local nonpartisan races moved onto the long even-year ballot, how many returned ballots would skip them? It is estimated from certified 2024 precinct returns (`scripts/diag_wa_rolloff_2024.py`), as (ballots counted − votes cast in a contest) / ballots counted (official 2024 ballots = 3,961,569).

| Contest type (2024 general) | Example | Roll-off |
|---|---|---:|
| Top of ticket | President | 1.1% |
| Partisan statewide office | Governor … Insurance Comm. | 2.7–6.8% |
| Statewide ballot measure | I-2109 … I-2124 | 4.1–5.6% |
| **Nonpartisan statewide, contested** | Supreme Court Pos. 2; Supt. of Public Instruction | **16.6–17.2%** |
| **Nonpartisan statewide, uncontested** | Supreme Court Pos. 8, 9 | **33.7–34.4%** |

Court of Appeals (regional; only one division votes) and Lt. Governor (loaded in only 5,355 of 8,111 precincts / 38 of 39 counties, a partial-load artifact, not roll-off) are excluded; see the script header.

Two honest points follow. First, Washington's even-year nonpartisan roll-off is not trivial: about **17%** in contested statewide nonpartisan contests and about **34%** in uncontested ones in 2024. These are plausible upper-bound analogs for local nonpartisan races, not direct estimates of city-council or school-board roll-off; voters generally have less information about judicial and local nonpartisan contests than about partisan statewide offices, and the age profile of Washington local-contest roll-off remains untested.

Second, even under large roll-off assumptions consolidation would probably enlarge the contest-level electorate but the *size* of that enlargement depends on whether local races move onto a **presidential-** or a **midterm-year** ballot. Using Washington's recent official even-year turnout as the baseline (share of registered casting a vote in the local contest):

| Scenario (baseline turnout) | 5% roll-off | 17% roll-off | 34% roll-off |
|---|---:|---:|---:|
| Presidential-year, 2024 (~79%) | ~75% | ~66% | ~52% |
| Midterm-year, 2022 (~64%) | ~61% | ~53% | ~42% |
| Odd-year, 2021/23/25 avg (~38%), current baseline | — | — | — |

The enlargement claim is strongest for presidential-year consolidation, still plausible for midterm-year consolidation, and weakest for low-salience or uncontested local races placed on a *midterm* ballot (~42% vs the ~38% off-year baseline). The grid also treats roll-off as *fixed* across scenarios, which it is not: a presidential electorate contains more peripheral voters, people who turn out only for the highest-salience contest and skip down-ballot races at higher rates, so its true roll-off is likely higher than a midterm electorate's, making the presidential row an upper bound and the presidential–midterm gap narrower than shown. The better conclusion is not that roll-off is second-order in every case, but that even substantial roll-off does not erase the turnout advantage unless local-contest roll-off runs far higher than the contested-nonpartisan benchmark.

**Does roll-off skew young? An ecological first look.** The concern behind this objection is that the young voters consolidation would add are also the ones most likely to skip a local contest. Cast-vote records cannot answer this directly as ballots are anonymous and carry no voter age, so the roll-off *age profile* is unmeasurable at the individual level under secret-ballot rules, and an ecological cut is the ceiling. Across the 39 counties, roll-off on that contest is if anything *higher* where the electorate is older (a correlation, Pearson's r, the standard linear correlation coefficient, of **+0.57**, ≈+0.6, **uncorrected for urbanicity**); that is, the county pattern does not show younger places skipping the contest more. This is weak evidence: it is ecological, confounded with the rural/urban gradient (older counties are rural, with thinner coverage of statewide judicial races), and measured on a *statewide* judicial race that is an imperfect analog for hyperlocal contests. It points away from young-concentrated roll-off but cannot establish individual behavior. A finer precinct-level ecological analysis is the natural sequel, and the next paragraph runs it; an individual-level administrative test is not available from public cast-vote records, because ballot secrecy prevents linking contest choices to voter age.

**A closer look, precinct by precinct.** The county number has one real weakness: in Washington the older counties are also the rural ones, so a county-level correlation cannot tell whether roll-off follows *age* or just *rural*. The +0.6 blends the two; the earlier county cut said as much. Precincts let us separate them. Across the ~4,900 precincts that have both a 2024 presidential vote and Census demographics ( `scripts/diag_wa_rolloff_precinct.py` ), roll-off in the Supreme Court Pos. 2 race averages 16.4%, in line with the roughly 17% we see statewide once both are measured the same way, a good check that the precinct data holds together. The county cut's *direction* survives; its *strength* does not. On the same yardstick the county used (the share of a precinct's 2024 voters who are 65 or older), the correlation falls from +0.6 to about +0.09, under a sixth as strong, because moving from 39 counties down to thousands of precincts strips out the lumping-together that had exaggerated it. (Measured against a precinct's older *residents* instead of its older *voters*, it is about +0.26, which is small but pointing the same way.)

The real value of working at the precinct level is that we can account for how urban and how well-off a place is. Put the age figures next to each precinct's income, education, home values, share of renters, and size as a stand-in for the urban-versus-rural and rich-versus-poor divide, and ask what age still explains on its own. Under that analysis almost nothing is left, with a correlation of only about +0.11 in the contested court race and +0.02 for Superintendent. So the big county number does not hold up either way and the little that remains still points the *opposite* way from the worry: if anything, older precincts skip the race slightly *more*, not less. The takeaway is the same as the county version, just on firmer ground: nothing in these ecological checks supports the fear that the young voters consolidation would bring in are the ones who would skip a local race. The usual cautions still apply, and they stack up rather than fade: this is a neighborhood-level pattern, so roll-off by a voter's own age can never be measured directly; the urban/rural stand-in is rough, since precincts are drawn to hold about the same number of people and a real density measure from the map files would do better; and a statewide court race is only a loose match for a city-council or school-board contest. But every test comes out the same way. (This precinct cut leans on two tables: Census figures mapped onto precincts, and a voter-file-to-precinct crosswalk; see the script header.)

## Appendix G — Off-cycle drop-off by precinct race, income, and education (ecological)

Appendix G is exploratory and is not used to carry the paper's main finding. The body of this paper measures age because the voter file carries each voter's birth year. It carries no race, income, or education (Washington does not publish them), so those can only be looked at *ecologically*, through the Census make-up of a voter's precinct, with the same ceiling as Appendix F: a precinct-level pattern is not proof about individuals.

With that caveat, the question is whether the precincts that drop off most between a presidential and an off-year are also the more nonwhite, lower-income, or less-college ones. Using off-cycle *retention*, the share of a precinct's 2024 presidential voters who came back for the 2025 off-year ( `scripts/diag_wa_offcycle_dropoff_demographics.py` , ~4,700 precincts), the raw picture is what one would expect: whiter, more-college, older precincts hold onto more of their voters off-cycle (Pearson $r \approx +0.25$ on % white, +0.18 on % college, +0.27 on the 65+ share), more-Hispanic precincts hold onto fewer (−0.20), and income is nearly flat (−0.08).

The sharper question is whether any of that survives the age story, since older precincts are also whiter. Holding the precinct's 65+ share constant, **education stays the strongest** as more-college precincts retain more voters off-cycle regardless of age (partial $r \approx +0.20$) while race attenuates but does not vanish (+0.10 on % white, −0.12 on % Hispanic) and income stays near zero. These are ecological patterns about places, not individual-level estimates about voters. They point to a representation gap worth studying more directly, not to a settled race- or education-level voter finding.

Every caveat from Appendix F applies and then some. This describes precincts, not people; it cannot show that any individual nonwhite or less-educated voter is likelier to skip an off-year. Retention is measured off the current voter file (survivorship applies), the precinct demographics are apportioned ACS estimates, and race in particular is unmeasurable at the individual level in Washington at any geography, because the voter file has no race field. It points to a gap worth a dedicated, better-controlled study, not a settled finding.

## Appendix H — The age gradient, one year at a time

The body of this paper reports composition in conventional age bands (the Census reporting brackets, plus the 18–29 and 65+ lines the turnout literature uses). Bands are a presentation choice, and a fair question is whether any banding choice manufactures, or hides, the result. Because the voter file carries every voter's year of birth, the question can be answered directly: this appendix drops the banding entirely and reports the gradient one year of age at a time ( `scripts/diag_wa_age_curve.py` ; re-derived by `verify_who_decides_wa.py` #30).

For each single year of age from 18 to 95, the table shows the April 2026 roll count, participation in the November 2024 presidential general and the November 2025 off-year general (current-roll reconstructions, with the same caveats as the body's rate table), and off-year **retention**: the share of that age's 2024 voters who returned a ballot in 2025. Retention is the cleanest of the three measures here, because both events are read off the same roll, so the current-roll denominator largely cancels.

| Age | Roll | 2024 turnout | 2025 turnout | Retention |
|---:|---:|---:|---:|---:|
| 20 | 75,896 | 55.3% | 14.2% | 23.0% |
| 25 | 91,312 | 52.1% | 15.5% | 26.3% |
| 30 | 96,668 | 55.9% | 20.2% | 32.9% |
| 35 | 106,887 | 61.9% | 25.4% | 38.4% |
| 40 | 99,309 | 67.5% | 29.4% | 41.6% |
| 45 | 92,301 | 71.2% | 32.3% | 43.6% |
| 50 | 78,068 | 74.0% | 34.8% | 45.5% |
| 55 | 86,879 | 77.0% | 38.0% | 47.9% |
| 60 | 78,167 | 79.0% | 42.1% | 51.8% |
| 65 | 84,252 | 82.7% | 49.7% | 58.6% |
| 70 | 77,479 | 86.3% | 58.4% | 66.3% |
| 75 | 62,085 | 88.2% | 63.1% | 70.5% |
| 80 | 39,097 | 88.5% | 64.2% | 71.4% |
| 85 | 22,346 | 85.9% | 60.3% | 68.9% |
| 90 | 10,165 | 82.2% | 54.3% | 64.6% |

Five-year steps shown; the script prints all 78 single ages. Ages 18–19 are omitted from the table: an 18-year-old in November 2025 was 17 at the November 2024 general, so that retention cell is an empty-denominator artifact, and 19-year-olds (23.4% retention) sit on the first-election pattern described below.

Four features of the full curve matter for reading the body of the paper.

**First, the gradient is a smooth age ramp, not a set of cohort steps.** Retention rises steadily from 23.0% at age 20 to a peak of 72.0% at age 79, at roughly one to one and a half points per year of age, with no visible breakpoint at the conventional cohort boundaries used in the body of the paper and no birth-year cohort standing out from its neighbors. That pattern shows that the main result is not manufactured by the choice of age bands. It is also consistent with a life-cycle interpretation, but it does not prove one: a durable high-turnout cohort currently concentrated in older ages could produce a similar cross-section, and separating age from cohort requires a longer panel.

**Second, there is no discontinuity at 65.** If the senior tilt reflected something categorical about retirement (time freed for civic life, or a benefits-driven interest spike), the curve should step upward near 65. It does not: retention moves from 57.5% at 64 to 60.5% at 66, a two-year step of 3.0 points, against a 5.7-point step across the four years from 60 to 64. The slope through the mid-60s is the same slope as the rest of the ramp. "65+" is a reporting convention, not a behavioral boundary.

**Third, the only non-monotonic features sit at the two tails.** At the young end, turnout shows the familiar first-election pattern: 18-to-20-year-olds participate in a presidential year at slightly *higher* rates (55.3% at 20) than voters in their mid-20s (52.1% at 25), then drop into the early-20s trough before the long climb begins. At the old end, the curve peaks in the late 70s and declines from about age 84 (retention 69.8% at 84, 64.6% at 90), consistent with health and mortality effects. The body's 75+ band blends the plateau and this decline.

**Fourth, the composition findings are therefore robust to any banding.** A monotone curve means every possible cut of the age axis preserves the ordering the paper reports; no alternative bracket scheme could reverse it. It also means data-driven clustering (for example, k-means on the age axis) would not recover natural behavioral cohorts: with no internal breakpoints, such methods would split the curve at densities of the *roll* (the largest birth cohorts), not at changes in behavior. The single-year curve is also the most suggestive evidence in this paper for a **life-cycle** interpretation of the senior tilt, subject to the age-cohort-period limit stated above and in the body's limits section. The habit-formation literature (Plutzer 2002, in Appendix D) points the same way.

## End note — data, reproduction, and series

**Data.** Washington's statewide voter-registration database (April 2026 extract): the 5.51M-voter roll joined to 27.1M individual vote records and each voter's year of birth (`data/wa_vrdb.duckdb`); access terms in Appendix B. Official ballot counts are the certified statewide totals published by the WA Secretary of State (`results.vote.wa.gov`, per-election `turnout.html` pages). Adult-resident and citizen-voting-age composition are the U.S. Census American Community Survey 2020–2024 5-year, tables B01001 (sex by age) and B29001 (citizen voting-age population by age), Washington (FIPS 53), retrieved through the Census API and reproduced by `scripts/acs_wa_adult_age.py`.

**Institutional context.** Washington is an unusually informative case because the formal administrative cost of voting is lower than in many states: registered voters are mailed a ballot, which they can return by mail without postage or drop in a box; eligible voters can register or update their registration in person up to 8 p.m. on Election Day; and registration is automatic through qualifying agency transactions (WA Secretary of State). Those rules do not remove every barrier as information costs, mobility, local attachment, address stability, and uneven political recruitment are still factors, but they make it hard to chalk the age gap up mainly to the friction of voting.

**Reproduction.** `scripts/verify_who_decides_wa.py` re-derives every count in the tables above from scratch (sections #1–#30, incl. the county breakdown, habitual-core overlap, snapshot cross-validation, gender, representativeness index, and the single-year-of-age curve of Appendix H, computed by `scripts/diag_wa_age_curve.py`); the roll-off appendix and its ecological age correlation are in `scripts/diag_wa_rolloff_2024.py`, with the finer precinct-level, SES- and urban-proxy-adjusted cut in `scripts/diag_wa_rolloff_precinct.py`; `scripts/diag_wa_individual_findings.py` and `scripts/diag_turnout_decomposition.py` produce the underlying figures; and `scripts/acs_wa_adult_age.py` reproduces the adult-resident and CVAP rows from the Census API. All scripts, the paper source, and the data-acquisition recipe are public at https://github.com/skirby359/who-decides.

**Series.** Lead paper of the electoral-health series (with `electoral-health-whitepaper.md` and `cross-state-fec-money.md`). Party-resolved companions in states that publish party of record: `who-decides-new-york.md` (deep blue) and `who-decides-idaho.md` (deep red).

**Companion paper: Safe-Seat Washington.** Once in an off-year general, how often is the contest even a choice? It counts **observed** competitiveness of every partisan legislative + congressional seat, 2016–2024 (≈85% non-competitive), and extends the count to a four-state lower-chamber map.